ual reviews have become increasingly challenging. Researchers now rely on advanced tools such as autonomous agents, to navigate vast datasets of scientific literature efficiently. These technologies facilitate automatic extraction of relevant information, trend analysis, and predictive modeling, accelerating the rate of discovery.

Agentic AI systems have the potential to address these challenges by automating information retrieval, extraction, and synthesis. However, automating literature review is a complex task that requires advanced natural language understanding, domain-specific knowledge, and the ability to handle ambiguity and nuance. Several frameworks have been developed to automate or augment the literature review process using Agentic AI. SciLitLLM Li et al. (2024c) is a proposed framework designed to enhance the scientific literature understanding. It employs a hybrid strategy that combines continual pre-training (CPT) and supervised fine-tuning (SFT) to infuse domain-specific knowledge and improve instruction-following abilities. While SciLitLLM demonstrates improved performance on tasks such as document classification, summarization, and question answering, making it a valuable tool for literature review, the framework relies heavily on high-quality training data, which may not always be available for emerging fields. Ajith et al. (2024) introduced LitSearch, a benchmark designed to evaluate retrieval systems on complex literature search queries in machine learning and NLP. The main strengths of LitSearch is its ability to provide a standardized framework for assessing the performance of retrieval systems, enabling researchers to compare different approaches and identify areas for improvement. In contrary, the benchmark is limited to specific domains (ML and NLP), which may restrict its applicability to other fields. ResearchArena Kang & Xiong (2024) is a benchmark for evaluating LLM-based agents in academic surveys, dividing the process into three stages: information discovery, selection, and organization. It helps assess AI performance in structured literature reviews but struggles to capture the complexity of real-world reviews. CiteME Press et al. (2024) evaluates language models' ability to accurately attribute scientific claims to their sources, focusing on machine learning literature. While CiteME addresses a crucial aspect of literature review by ensuring accurate citation, it is limited in scope, restricting its application to other fields.

Despite the progress made by existing frameworks, several challenges remain in automating the literature review process. While frameworks such as SciLitLLM and ResearchArena demonstrate promising results, they often struggle with tasks requiring deep domain-specific knowledge and nuanced understanding. This limitation is further highlighted in Agent Laboratory Schmidgall et al. (2025), where a significant performance drop was observed during the literature review phase, emphasizing the complexity of automating this process. Another challenge lies in human-AI collaboration, as many existing frameworks prioritize fully autonomous workflows. This approach may limit usability for researchers who want to explore their unique ideas, underscoring the need for collaborative approaches that effectively integrate human expertise with AI capabilities. Generalizability is also a major obstacle, as many frameworks are designed for specific domains like machine learning, chemistry, or materials science, which restricts their application in other fields.

5 Agentic AI for Scientific Discovery

Agentic AI systems are revolutionizing the scientific research process by automating and augmenting various stages of the research lifecycle, from ideation and experimentation to paper writing and dissemination. Figure 1 depicts the agentic AI workflow for scientific discovery. These systems leverage the capabilities of LLMs and other AI technologies to streamline workflows, reduce human effort, and accelerate the pace of discovery. In this section, we explore how Agentic AI is transforming scientific discovery, supported by case studies and a discussion of key challenges. The research lifecycle traditionally involves several stages, including problem identification, literature review, hypothesis generation, experiment design, data analysis, and publication. Agentic AI systems are being deployed to automate or augment each of these stages, enabling researchers to focus on high-level decision-making and creative problem-solving. Here are the main steps:

- **Ideation:** Ideation refers to the process of generating, refining, and selecting research ideas or hypotheses. AI agents automate this process by analyzing existing literature, identifying gaps, and proposing novel hypotheses, thereby accelerating the initial stages of research Baek et al. (2024).

- **Experiment design and execution:** Experiment design involves planning and structuring experiments to test hypotheses, while execution refers to carrying out these experiments. AI agents autonomously design and execute complex experiments by integrating tools for planning, optimization, and robotic automation Boiko et al. (2023).
- **Data analysis and interpretation:** Data analysis involves analyzing experimental data to extract meaningful insights, while interpretation refers to drawing conclusions and identifying patterns. Agents can process large datasets and generate insights that might be overlooked by researchers, enhancing the accuracy and efficiency of this stage.
- **Paper writing and dissemination:** Paper writing involves synthesizing research findings into a coherent and structured manuscript, while dissemination refers to sharing the research with the scientific community through publications or presentations. AI agents automate the writing of research papers, ensuring clarity, coherence, and adherence to academic standards, thereby reducing the time and effort required for publication Lu et al. (2024).

By relying on LLM-augmented agents, these systems have made a significant strides in scientific discovery in domains such as chemistry, biology, materials science as well as general science where the main dream is to develop a fully autonomous AI scientist. In chemistry, AI agents are transforming key areas such as molecular discovery and design, reaction prediction, and synthesis planning by accelerating the identification of novel compounds and optimizing synthetic routes. Additionally, they contribute to laboratory automation, integrating with robotic systems to execute experiments autonomously, and enhance computational chemistry by running molecular simulations for reaction kinetics and thermodynamics.

Coscientist Boiko et al. (2023) is an autonomous AI agent powered by GPT-4 that plans, designs, and executes chemical experiments. It integrates modules for web search, documentation analysis, code execution, and robotic automation, enabling it to handle multi-step problem-solving and data-driven decision-making. For example, Coscientist successfully designed and optimized a palladium-catalyzed cross-coupling reaction, demonstrating its potential to accelerate chemical discovery. Similarly, Ruan et al. (2024) proposed LLM-RDF (Large Language Model Reaction Development Framework), a framework that automates chemical synthesis using six LLM-based agents for tasks like literature search, experimental design, reaction optimization, and data analysis. Tested on copper/TEMPO catalyzed aerobic alcohol oxidation, it demonstrates end-to-end synthesis automation. It simplifies reaction development, making it more accessible to chemists without coding expertise. In the same context, Chiang et al. (2024) developed a novel framework called LLaMP (Large Language Model Made Powerful), designed for scientific discovery in chemistry by integrating RAG with hierarchical reasoning agents. It significantly reduces hallucination in material informatics by grounding predictions in high-fidelity datasets from the Materials Project (MP) and running atomistic simulations. LLaMP successfully retrieves and predicts key material properties such as bulk modulus, formation energy, and electronic bandgap, outperforming standard LLMs. Darvish et al. (2024) introduced Organa, an assistive robotic system designed for automating diverse chemistry experiments, including solubility screening, pH measurement, recrystallization, and electrochemistry characterization. Using LLMs for reasoning and planning, Organa interacts with chemists in natural language to derive experiment goals and execute multi-step tasks with parallel execution capabilities. In electrochemistry, it demonstrated the automation of complex processes, such as electrode polishing and redox potential measurement, achieving results comparable to human chemists while reducing execution time by over 20%. This system enhances scientific discovery by improving the reproducibility and efficiency of chemistry experiments. Other notable frameworks based on AI agents for scientific discovery in chemistry include ChatMOF Kang & Kim (2023), ChemCrow M. Bran et al. (2024) and MOOSE-CHEM Yang et al. (2024) among others.

In addition to chemistry, AI agents, powered by LLMs and multi-agent systems, are transforming biology by enabling automated data analysis, hypothesis generation, and experimental planning. These agents can extract insights from vast amounts of biological data, such as genomic sequences, protein structures, and biomedical literature, to accelerate research across fields like genetics, drug discovery, and synthetic biology. With capabilities such as gene-editing design, protein engineering, and systems biology modeling, AI agents are playing a critical role in scientific discovery for biology. By integrating with laboratory tools and robotic systems, they not only reduce human effort but also enhance research accuracy and reproducibility, bringing us closer to breakthroughs in personalized medicine, disease modeling, and bioinformatics. Xin et al. (2024) introduced BIA

(BioInformatics Agent), an AI agent leveraging LLMs to streamline bioinformatics workflows, particularly focusing on single-cell RNA sequencing (scRNA-seq) data analysis. BIA automates complex tasks like data retrieval, metadata extraction, and workflow generation, significantly improving bioinformatics research efficiency. It features a chat-based interface for designing experimental protocols, invoking bioinformatics tools, and generating comprehensive analytical reports without coding. BIA's innovative use of static and dynamic workflow adaptation allows it to refine bioinformatics analyses iteratively, demonstrating its potential to reduce the cognitive load on researchers and enhance scientific discovery in genomics and transcriptomics. Similarly, Xiao et al. (2024) developed CellAgent, an LLM-driven multi-agent system designed to automate single-cell RNA sequencing data analysis. It features three expert roles: Planner, Executor, and Evaluator, which collaborate to plan, execute, and evaluate data analysis tasks such as batch correction, cell type annotation, and trajectory inference. CellAgent reduces human intervention by incorporating a self-iterative optimization mechanism, achieving a 92% task completion rate and outperforming other scRNA-seq tools in accuracy and reliability. This framework significantly enhances biological research efficiency, making scRNA-seq analysis accessible to non-experts and enabling new biological discoveries. Liu et al. (2024) developed TAIS (Team of AI-made Scientists), a semi-autonomous AI assistant for genetic research, designed to suggest and refine biological experiments using self-learning mechanisms. It helps scientists with data analysis, hypothesis generation, and experiment planning, but requires human validation before execution. Other works include ProtAgents Ghafarollahi & Buehler (2024a), AI Scientists Gao et al. (2024) and CRISPR-GPT Huang et al. (2024a).

In addition to the previous domains, AI agents are widely explored in other fields such as materials science Ni et al. (2024); Maqsood et al. (2024); Papadimitriou et al. (2024); Strieth-Kalthoff et al. (2024); Merchant et al. (2023); Kumbhar et al. (2025), general science Taylor et al. (2022); Yang et al. (2023b); Baek et al. (2024); Lu et al. (2024); Swanson et al. (2024); Qi et al. (2023); Ghafarollahi & Buehler (2024b); Schmidgall et al. (2025) as well as machine learning Li et al. (2024b); Huang et al. (2024b); Chan et al. among others. Figure 2 shows the summary of these agentic AI frameworks for scientific discovery in various domains.

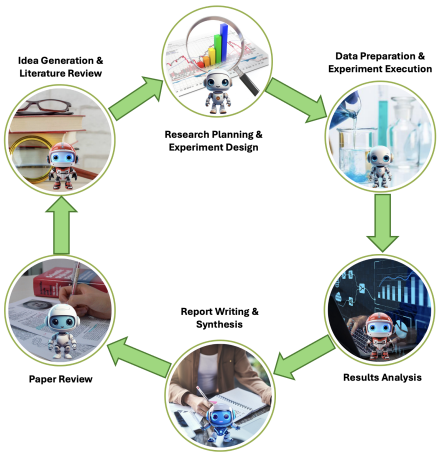

Figure 1: Agentic AI workflow for scientific discovery.

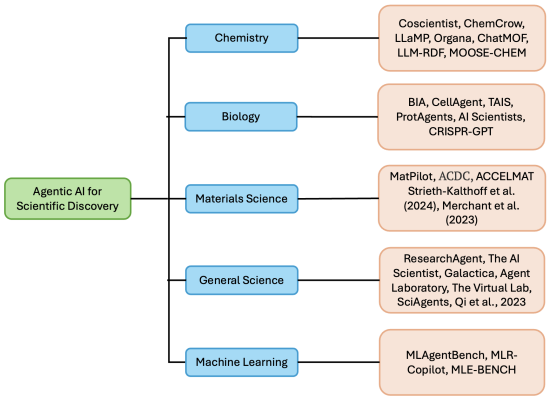

Figure 2: AI Agents frameworks for scientific discovery.

6 Implementation Tools, Datasets and Metrics

The development and evaluation of agentic AI systems for scientific discovery rely on a robust tools, curated datasets, and well-defined evaluation metrics. This section provides an overview of the key resources used in the field to facilitate the design, training, and assessment of autonomous AI agents for scientific discovery.